

Hemolytic Anemias - Factoid Sheet

Week 7 Hematology • Lecture 5 • Dalgetty • memorize-first companion

ScholarRx: [RBC Metabolism & Hemolytic Anemia](#) • [Hereditary Spherocytosis](#) • [MAHA](#) • [AIHA](#) • [Blood Transfusion](#)

Hemolysis signature: anemia + retic up, LDH up, indirect bilirubin up, haptoglobin down.

Intravascular

- RBC bursts in vessel
- Hemoglobinemia / hemoglobinuria
- Hemosiderinuria later
- **Very low haptoglobin**
- Mechanical shear, complement, toxin/infection

Extravascular

- Spleen/liver macrophage clearance
- Splenomegaly
- Indirect bilirubin / jaundice
- Spherocytes common
- HS, warm AIHA

Smear -> diagnosis

Finding	Think	Discriminator
Spherocyte	HS or warm AIHA	DAT negative = HS; DAT positive = warm AIHA
Bite cell	G6PD	Oxidant trigger; Heinz bodies
Schistocyte	MAHA/TTP/HUS/DIC	Platelets, coag tests, renal/neuro findings
Parasite	Malaria/babesia	Exposure/travel

Hereditary spherocytosis

- Spectrin/ankyrin/band 3-type membrane defect
- Loss of surface area -> spherocyte -> splenic trapping
- **MCHC high**
- Osmotic fragility increased (slide)
- Splenectomy reduces hemolysis; pigment gallstones

G6PD deficiency

- X-linked pattern
- Low NADPH -> low reduced glutathione -> oxidative injury
- Triggers: infection, oxidant meds, fava beans
- Heinz bodies -> bite cells
- **G6PD level may be falsely normal during attack**

Warm vs cold AIHA

	Warm	Cold agglutinin
Ab	IgG	IgM
Temp	37 C	Cooler temps
Mechanism	Splenic Fc removal -> spherocytes	Complement fixation
DAT	IgG +/- C3	C3d
Associations	SLE, CLL, drugs	Mycoplasma, lymphoid disease
Tx theme	Steroids / rituximab	Avoid cold; complement-directed therapy / treat cause

MAHA, TTP & Transfusion Reactions

Emergency rule: schistocytes + thrombocytopenia is not a "review later" finding. Evaluate TMA/TTP immediately.

TTP

Factoid	Must know
Mechanism	Severe ADAMTS13 deficiency -> ultralarge vWF multimers -> platelet-rich microthrombi -> RBC shear
Labs/smear	MAHA + schistocytes + thrombocytopenia; PT/aPTT often normal
Symptoms	Neuro symptoms common; renal findings can occur; fever possible
PLASMIC	6-7 high probability; 5 intermediate; 0-4 low in slide
Critical action	Do NOT wait for ADAMTS13. Start plasma exchange when strongly suspected.
Lecture treatment	PLEX + steroids + rituximab +/- caplacizumab

TTP vs DIC

TTP: platelet-VWF microthrombi; PT/aPTT often normal.
DIC: coagulation-factor consumption; PT/aPTT often prolonged, fibrinogen low, D-dimer high.

Coombs

Direct DAT Are Ig/complement already attached to the patient's RBCs? Use for AIHA/ transfusion-reaction evaluation.	Indirect Are anti-RBC antibodies floating in serum? Used in pretransfusion antibody screening.
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Hemolytic transfusion reactions

	Acute	Delayed
Timing	During / soon after transfusion	Days later (companion: ~3-10 d)
Cause	Usually ABO incompatibility / preformed antibody	Anamnestic response to minor antigen
Immune	Often IgM + complement	Usually IgG
Hemolysis	Intravascular	Extravascular
Clues	Fever, hypotension, flank/back pain, hemoglobinuria	Unexpected Hb drop, jaundice, subtle symptoms

Acute transfusion-reaction first move: STOP the transfusion. Preserve IV access, verify identity, notify the blood bank, and evaluate for hemolysis.

If you see X -> think Y

• Spherocytes + DAT negative + family hx -> HS • Spherocytes + DAT positive -> warm AIHA • Bite cells after TMP-SMX/fava/infection -> G6PD • Acrocyanosis after cold/Mycoplasma -> cold agglutinin disease	• Schistocytes + Plt 20K + confusion -> TTP • Normal PT/aPTT with MAHA -> supports TTP over DIC • Dark urine + hypotension during transfusion -> acute HTR • Jaundice + falling Hb days later -> delayed HTR
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Last 5 minutes: retic/LDH/bili up, haptoglobin down. HS = spherocyte, high MCHC, DAT-. Warm = IgG/DAT+. Cold = IgM/C3. G6PD = Heinz -> bite cells; test can be false-normal during crisis. TTP = ADAMTS13, schistocytes + low platelets, PLEX now.

Primary source: Dalgetty 51-slide Hemolytic Anemias deck. Official Week 7 LOs control emphasis. Transfusion-reaction details come from the uploaded companion because the slide deck does not develop LO 7.